

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 2

Error Analysis Task

COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 30%

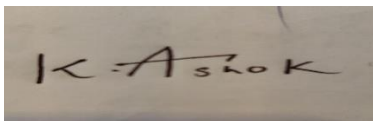
Duration: 30 minutes

Marks: 25

Code of Conduct:

I K Ashok(192425322) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

QUESTIONS:

Total Marks:25

1. Error Analysis in Maximum Likelihood Estimation (MLE)

A student builds a binary classifier using Maximum Likelihood Estimation. The likelihood function is incorrectly written as the sum of probabilities instead of the product. The model produces poor parameter estimates even with large data.

Tasks:

- Identify the conceptual error in the likelihood formulation.
- Explain how this mistake affects parameter estimation.
- Suggest the correct mathematical formulation and reasoning.
- How would using log-likelihood fix computational issues?

Machine Learning Error Analysis Answers

1. Error Analysis in Maximum Likelihood Estimation (MLE)

- The conceptual error is defining likelihood as a sum of probabilities instead of the product of probabilities for independent observations. MLE maximizes the joint probability of observing the entire dataset.
- Using a sum ignores the contribution of all observations jointly. The estimated parameters no longer maximize the probability of the complete dataset, leading to biased and inconsistent estimates even with large datasets.
- Correct formulation: $L(\theta) = \prod P(x_i | \theta)$. For independent samples, the joint likelihood is the product of individual probabilities because each observation contributes multiplicatively.
- Log-likelihood uses $\log L(\theta) = \sum \log P(x_i | \theta)$. It converts products into sums, prevents numerical underflow, simplifies differentiation, and makes optimization more stable.

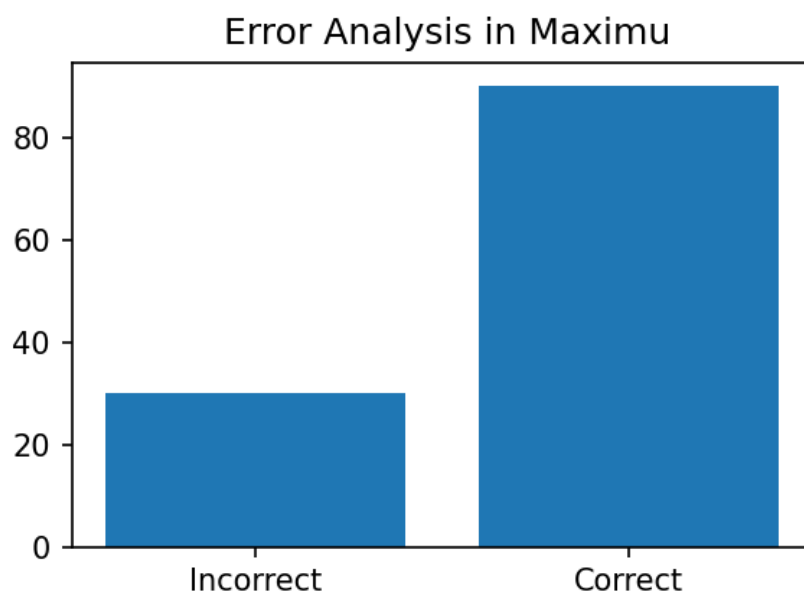


Figure 1: Illustrative bar chart related to the concept.

2. Error Diagnosis in Perceptron Learning

While implementing a Perceptron, a student observes that the model fails to converge even after many iterations on a linearly separable dataset.

Tasks:

- List possible implementation errors in the learning algorithm.
- Analyze how incorrect learning rate or weight updates affect convergence.
- Identify dataset-related issues that may cause this problem.
- Propose modifications to ensure convergence.

Error Diagnosis in Perceptron Learning

- Possible implementation errors include incorrect weight update rule, missing bias update, wrong prediction function, incorrect label encoding, or shuffled data not handled correctly.
- A very small learning rate slows convergence, while a very large rate causes oscillation. Incorrect weight updates prevent the decision boundary from moving correctly.
- Problems may include mislabeled samples, non-linearly separable data, missing bias term, duplicate noisy samples, or incorrect feature scaling.
- Use proper update rule, normalize features, include bias, verify labels (+1/-1), reduce learning rate if necessary, and ensure the dataset is linearly separable.

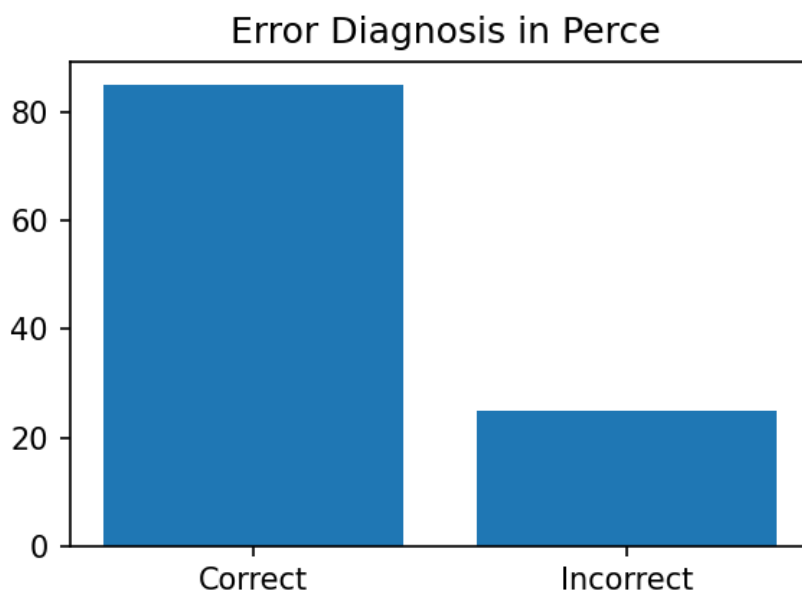


Figure 2: Illustrative bar chart related to the concept.

3. Backpropagation Gradient Error Analysis

A Multilayer Perceptron trained using Backpropagation shows increasing loss during training instead of decreasing.

Tasks:

- a) Identify possible mathematical or coding errors in gradient computation.
 - b) Explain the role of activation functions in gradient flow.
 - c) Analyze how vanishing or exploding gradients affect training.
 - d) Suggest techniques to fix the issue (e.g., normalization, initialization).
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- a)** Possible errors include incorrect chain rule implementation, wrong derivative of activation function, sign errors in gradient updates, or indexing mistakes.
 - b)** Activation functions control gradient flow. Sigmoid and tanh may saturate, while ReLU usually allows better gradient propagation.
 - c)** Vanishing gradients make updates extremely small, slowing learning. Exploding gradients make updates excessively large, causing unstable training and increasing loss.
 - d)** Use Xavier/He initialization, batch normalization, gradient clipping, ReLU-based activations, and adaptive optimizers such as Adam.

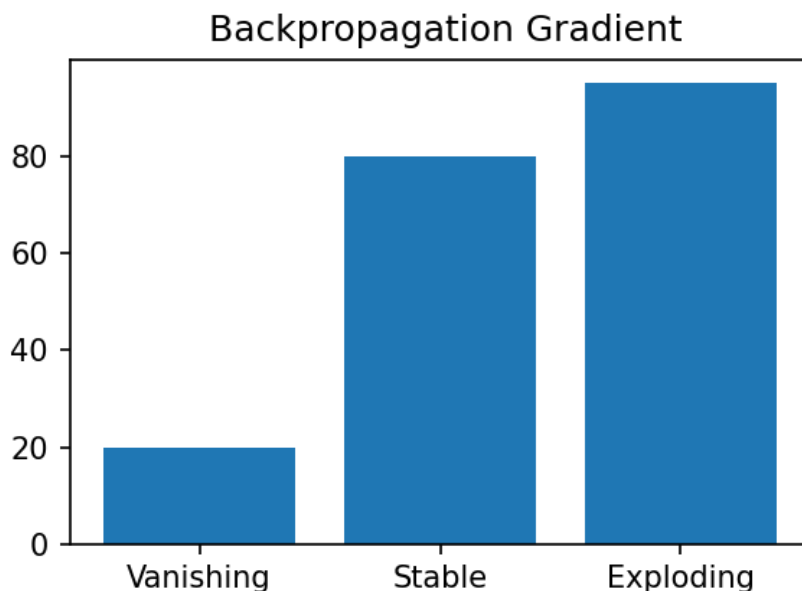


Figure 3: Illustrative bar chart related to the concept.

4. Gradient Descent vs SGD Error Investigation

A model trained using Gradient Descent converges very slowly, while the same model with Stochastic Gradient Descent shows unstable loss fluctuations.

Tasks:

- a) Explain why batch gradient descent is slow in large datasets.
- b) Analyze the cause of fluctuations in SGD.
- c) Compare the error behavior of Mini-Batch Gradient Descent.
- d) Recommend the best approach and justify with error trade-offs.

Answer:

- a) Batch Gradient Descent computes gradients using the entire dataset each iteration, making training slow on large datasets.
- b) SGD updates parameters after each sample. The noisy updates cause fluctuations but often help escape local minima.
- c) Mini-Batch Gradient Descent balances speed and stability by using small batches, reducing variance while remaining efficient.
- d) Mini-batch with Adam is generally the best practical choice because it provides faster convergence and more stable error behavior.

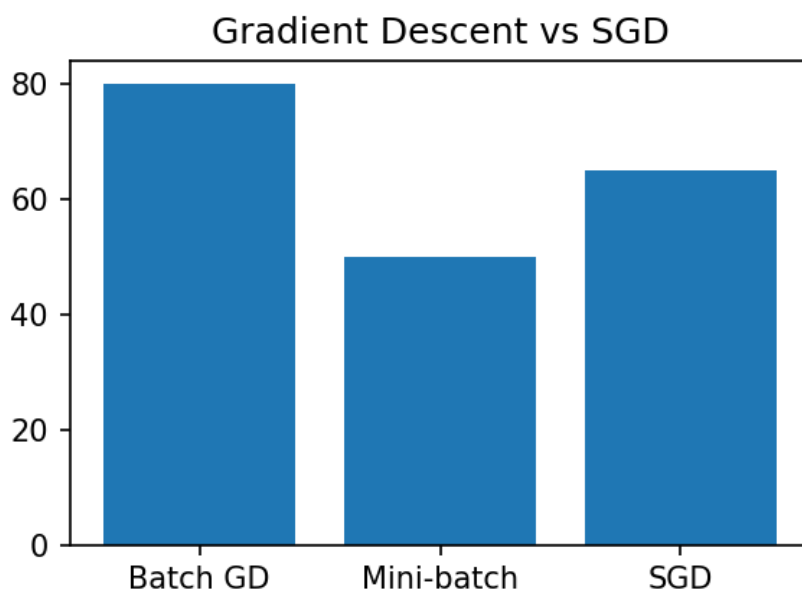


Figure 4: Illustrative bar chart related to the concept.

5. Curse of Dimensionality Error Analysis

A machine learning model built using high-dimensional data performs well on training data but poorly on test data due to overfitting, related to the Curse of Dimensionality.

Tasks:

- Explain how high dimensionality increases model error.
- Identify signs of overfitting in this scenario.
- Analyze why distance-based learning fails in high dimensions.
- Suggest dimensionality reduction or regularization techniques to improve performance.

Answer:

- As dimensions increase, data becomes sparse. Models memorize training data instead of learning general patterns, increasing generalization error.
- Signs include very high training accuracy, low test accuracy, large train-test performance gap, and unstable predictions.
- Distances between points become similar in high dimensions, reducing the usefulness of nearest-neighbor and clustering methods.
- Apply PCA, feature selection, L1/L2 regularization, dropout, or dimensionality reduction techniques such as t-SNE (visualization) or autoencoders.

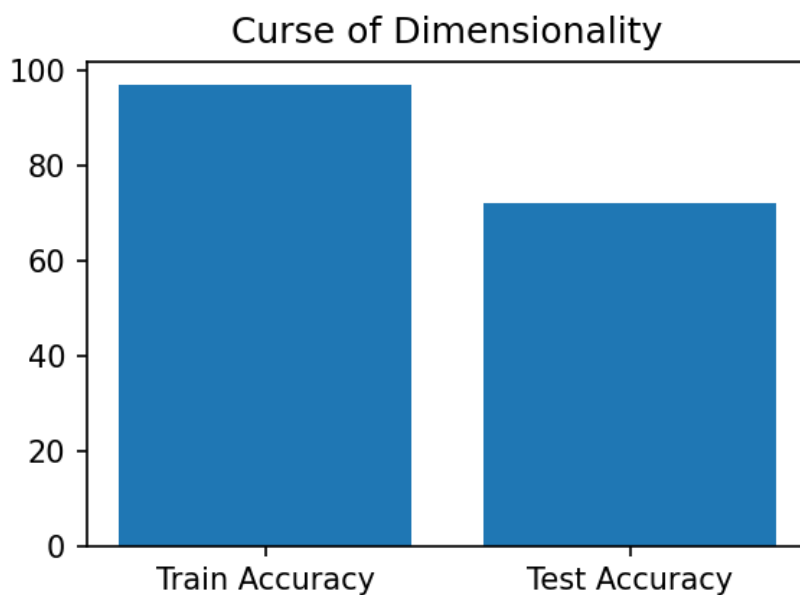


Figure 5: Illustrative bar chart related to the concept.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand